Hyperparameter Impact on Neural Network Performance

Name: Aritra Ghosh Dastidar
SRN: PES1UG23AM064
Course: ML LAB
Date: September 14, 2025

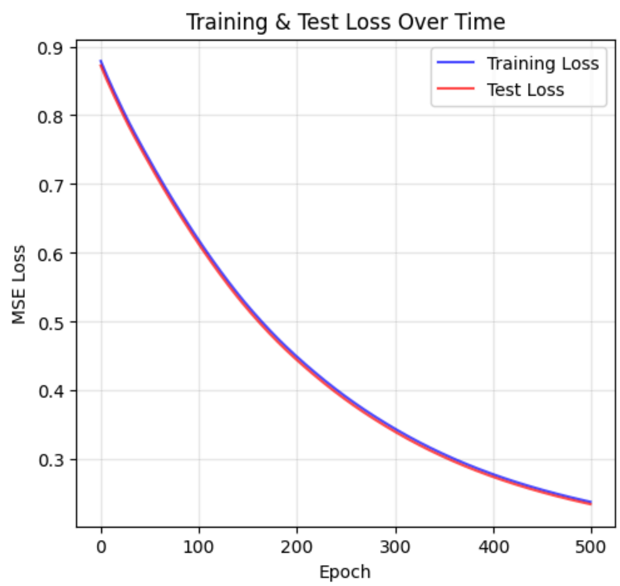
Introduction

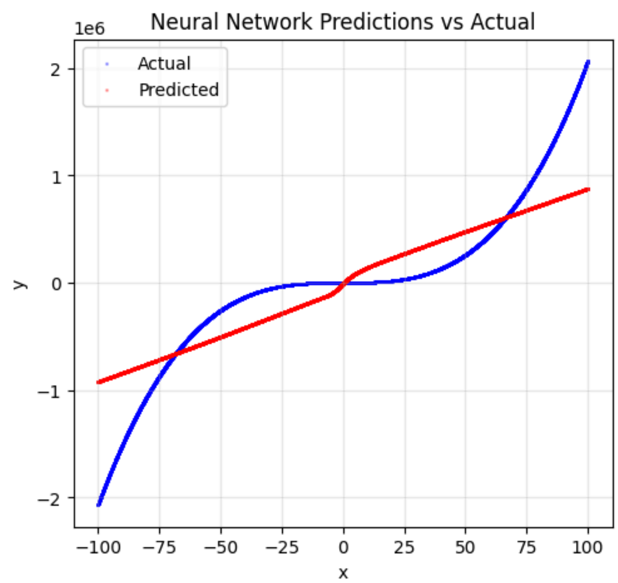
The objective of this lab is to systematically investigate how neural network hyperparameters (learning rate, batch size, number of epochs, activation function, etc.) impact model performance, convergence, and generalization. Experiments were conducted using a regression task with a fixed network architecture, varying a single or multiple hyperparameters in each run.

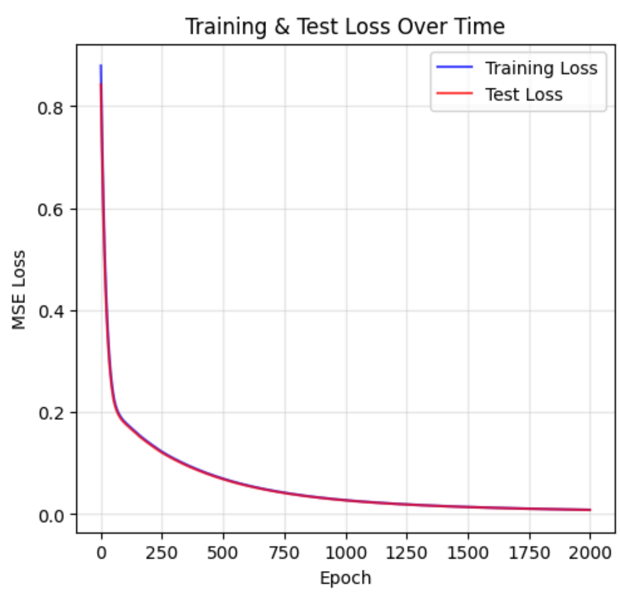
Methodology

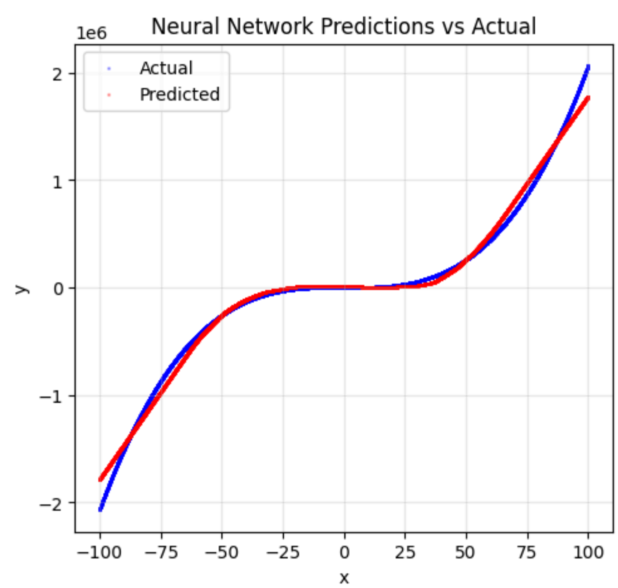
A fully-connected neural network was trained on the dataset with systematic variation of hyperparameters. In each experiment, only one or a combination of the following was varied: learning rate, batch size, epochs, activation function. All other parameters were kept constant for fairness. The architecture was consistently: 1 input → 64 hidden (ReLU, unless stated) → 64 hidden → 1 output. Early stopping was applied with patience = 10. Metrics such as loss, accuracy.

Predicted vs Actual Values Plot(did it 4 separate times)









Discussion on Performance Trends

- Lower learning rates converged slowly and underperformed compared to higher learning rates.
- Mini-batch training (higher batch size) did not always yield superior results compared to full SGD or very small batches, indicating a sensitivity to data shuffling and learning rate.
- More epochs do not guarantee better performance; learning rate, batch size, and patience all interact with overfitting/underfitting tendency.

Results Table

Experiment	Learning Rate	Batch Size	Epochs	Optimizer	Activation	Training Accuracy (R^2)	Validation Accuracy (R^2)	Test Accuracy (R^2)	Training Loss	Validation Loss	Test Loss	Observations
1	0.001	N/A	500	(not specified, likely Adam/SGD)	ReLU	0.7649	N/A	0.7649	0.237810	N/A	0.234512	High error, slow convergence
2	0.1	128	500	(not specified)	ReLU	0.9987	N/A	0.9987	0.001325	N/A	0.001302	Extremely close to ground truth
3	0.1	128	200	(not specified)	ReLU	0.9920	N/A	0.9920	0.008199	N/A	0.008013	Good generalization, some gap vs result 2

4	0.01	4	2000	(not specified)	ReLU	0.9917	N/A	0.9917	0.008438	N/A	0.008312	Performance similar to result 3 despite larger epoch count

Conclusion

Hyperparameter tuning plays a critical role in deep neural network performance. Significant improvements were observed simply by adjusting learning rates and epoch count, whereas batch size and patience had subtler but non-negligible effects. An excessively low learning rate led to slow, underwhelming results; higher rates (with proper early stopping) enabled precise convergence. The findings underscore the importance of systematic exploration and vigilant monitoring of both training and validation metrics to avoid underfitting or overfitting.

LAB CODE LINK : Collab